higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(Second Paper)
NQF LEVEL 2

25 FEBRUARY 2019

SYMBOL	EXPLANATION
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RD/RM	Reading from a table/graph/drawing/document/map
F	Choosing correct formula
SF	Substitution in formula
MF	Manipulation of formula
R/J	Reasoning/Justification
P	Penalty, e.g. for no units, incorrect rounding off etc.
R	Rounding off
E	Explanation

This marking guideline consists of 9 pages.

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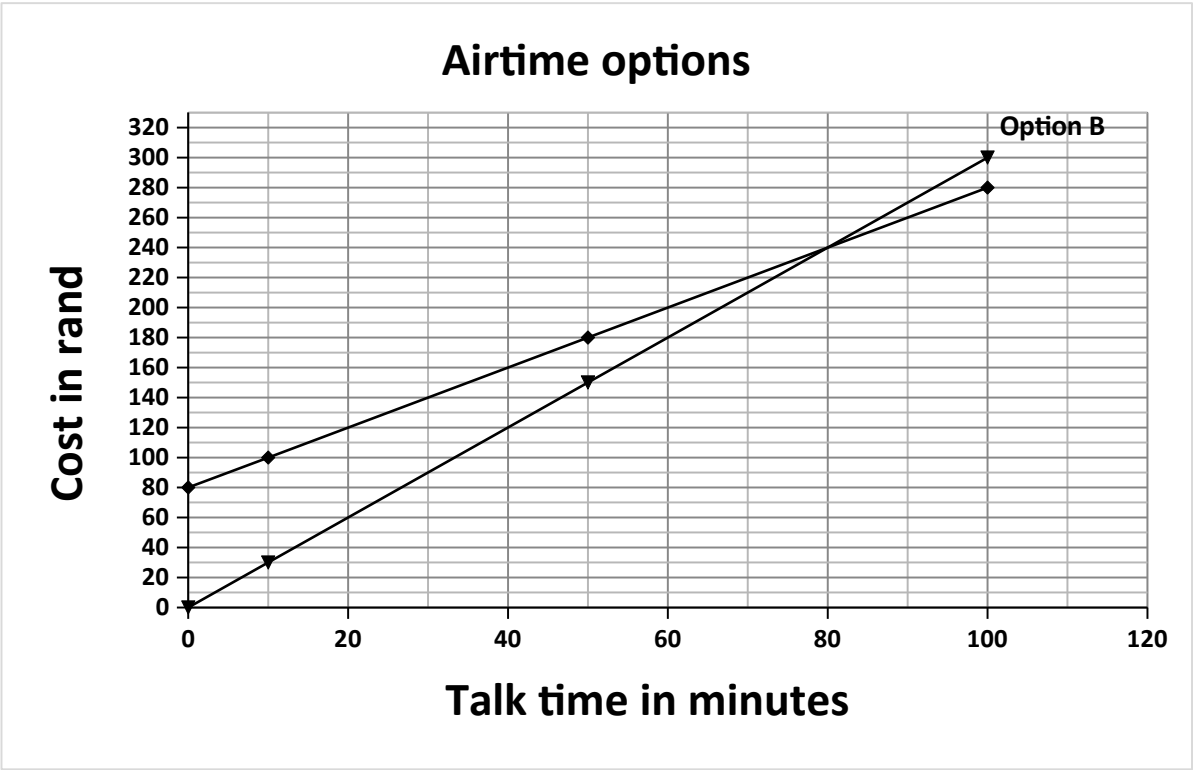
QUESTION 1	*** Penalise for unit only where indicated.**		
Question	**Solution**		**Explanation**
1.1	1.1.1	A – rectangle✓	1A (1)
	1.1.2	C – right angled triangle✓ (no mark for only triangle)	1A (1)
	1.1.3	Area of Paddock A = Length × breadth = 240 × 60✓ = 14 400 ✓m^2 Area of Paddock B = 14 400✓ m^2 Area of Paddock C = Area of triangle = ½×70✓ × 240✓ = 8 400 ✓m^2 Total area = 14 400 + 14 400 + 8 400✓ = 37 200✓ m^2 ✓	1SF 1A (2 Answer only) 1A 2SF (3 Answer only) 1A 1M (CA only with one mistake made in Area of A, B or C) 1A and 1 unit (9)
	1.1.4	$PR^2 = 70^2 + 240^2$ ✓ = 62 500✓ PR = $\sqrt{62\,500}$ ✓ = 250 ✓m	1SF 1A 1M 1A (4 Answer only) (4)
	1.1.5	Total Fencing = 240✓+60✓+240+240✓ +60+ 60 + 60✓ +70✓+250✓ = 1 280 m	1RD 240 1RD 60 1RD 240 1A 240 + 240 1A 60+ 60 + 60 1RD 70 1 CA 250 (Q1.1.4) (6 Answer only) (6)
	1.1.6	Cost of fencing = 1 280 ✓× R45✓ = R57 600✓ (No marks to be deducted for R omitted)	2 M 1 A (3 Answer only) (3)
	1.1.7	Map distance = 27km ✓÷ 500 000✓ = 0,000054✓ km = 0,000054✓ × 1 000 000✓ = 54 ✓mm	2MA 1A 2M 1CA (6 Answer only) (6)

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1.2	1.2.1	Volume = 10✓ × 10✓ × 45✓ = 4 500✓ cm^3✓	3SF 1A and 1 unit (5 Answer only) (5)
	1.2.2	Radius = 5 ✓cm Volume = 3,14✓ × 5^2✓ × 45✓ = 3 532,5 ✓cm^3 Accept: 3534,292 / 3534,29 / 3534,3	1SF 3,14 1SF r = 5cm 1SF h = 45 1CA (5 Answer only) (5)
			[40]

QUESTION 2 *Do not penalise if 'R' is not indicated			
Question	**Solution**		**Explanation**
2.1	2.1.1	Expenditure = 1 800 + 200 + 1 300 + 200 + 440 + 500 + 300 + 430+ 300 + 1225 + 400 + 2300 + 150 + 150✓ = R9 695✓	1MA - addition 1A (2 Answer only) (2)
	2.1.2	Surplus ✓= 9 956,55✓ – 9 695✓ = R261, 55	2M 1 for surplus (3)
	2.1.3	Groceries, Toiletries, Entertainment, Petrol, Clothing, cellphone, Water and electricity. (Any TWO) ✓✓	2 RT (2)
	2.1.4	Rent, School fees, Car repayment, cellphone, medical aid, Car insurance (Any TWO) ✓✓	2 RT (2)
	2.1.5	No,✓ the surplus is too little in case of an emergency✓✓ Or No,✓ she only saves R150 and this is too little, the money left is also too little especially in the case of emergency or unexpected expenses. (Any relevant reasoning) ✓✓	1 S 2 J (3)
	2.1.6	She can add it to her savings✓ and save at least R411,55 which will be a better saving than R150✓.	2 J (2)

	2.1.7	10% of 440 = 44✓ 10% of 1 225 = 122,50✓ 10% of 2 300 = 230✓ Total increase = R44 + R122,50 + 230 = R396,50✓ Total expenditure = R396,50 + R9 695 = R10 091,50✓ Surplus/Deficit = R9 956, 55 – R10 091,50 = R134,95✓ (Deficit) ✓ (Accept –R134,95 for one mark)	3MA 1A 1M (CA 2.1.1) 1A and 1 deficit (7)
2.2	2.2.1	Summary of the income which shows the deductions and the remaining money✓ ✓ (net). (Accept other appropriate definitions)	2 E (2)
	2.2.2	Gross salary = R7 998 + R875✓ = R8 873✓	1M 1A (2 Answer only) (2)
	2.2.3	Pay as you earn ✓	1 A (1)
	2.2.4	Percentage PAYE = $\frac{2218,25}{8873}$✓$\times 100$✓ = 25✓%	2M (CA 2.2.2) 1A (3 Answer only) (3)
	2.2.5	Total deductions = R4 047,96✓ Net salary = R8 873✓ - R4 047,96✓ = R4 825,04✓	1 A 2M 1CA (Q 2.2.2) (4)
2.3	2.3.1	Wheel alignment and balancing✓	1RT (1)
	2.3.2	VAT = $\frac{15}{100}$ ✓$\times 250,80$✓ = R37,62✓	1M, 1RT 1 A (3 Answer only) (3)
	2.3.3	Total amount = R250,80 ✓+ R37,62✓ = R288,42✓	2M (CA 2.3.2) 1CA(3 Answer only) (3)
			[40]

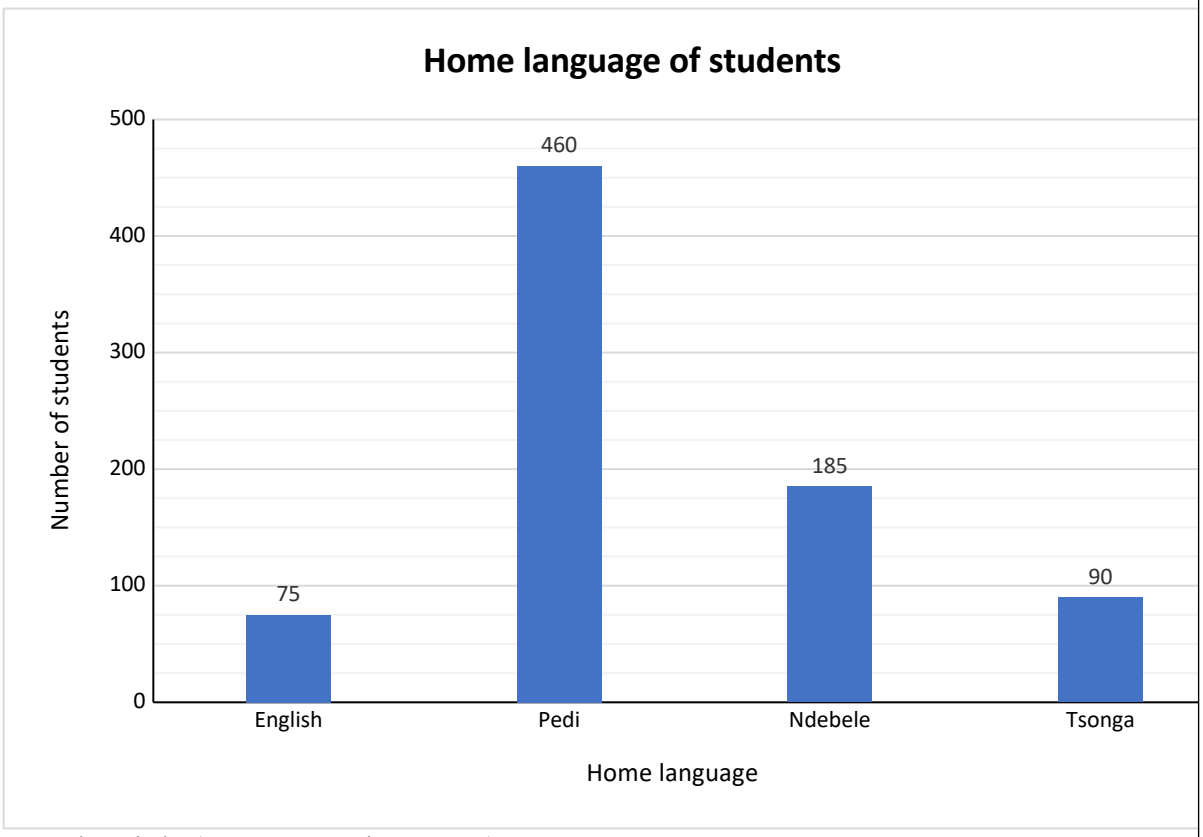
QUESTION 3 ***Do not penalise if *R* is not indicated**				
Question	**Solution**			**Explanation**
3.1	3.1.1	(a)	A = 0✓✓	2MA
		(b)	Total cost = R80 + R2,00 ¿ 50✓ B = R180 ✓	1M 1A (2 Answer only) (4)
	3.1.2	(a)	C = R3,00 ¿ 10✓ = R30✓	1M 1A (2 Answer only)
		(b)	D = 100✓✓	2MA (4)
	3.1.3		 1 Mark for title. 1 Mark for naming x axis 1 mark for naming y-axis Option A 4 Marks for correct plotting 1 Mark for straight line Option B 4 Marks for correct plotting 1 Mark for straight line (13)	

	3.1.4	Difference = 200 – 180✓ = R20✓	1RG 1CA(2 answer only) (2)
	3.1.5	80 ✓✓minutes	2RG (2)
3.2	3.2.1	As the number of trips increases ✓the cost also increase in the same proportion✓	2E (2)
	3.2.2	Relative cost per trip✓✓ (Accept other relevant headings)	2 A (2)
	3.2.3	R44✓✓	2RT (2)
	3.2.4	5 ✓✓trips	2RT (2)
	3.2.5	Relative cost per trip = R440,00 ÷25✓ = R17,60✓	1SF 1A(2 Answer only) (2)
			[35]

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QUESTION 4			
Question	**Solution**		**Explanation**
4.1	4.1.1	220✓+50+10✓+ 30✓+70 = 380✓	3RG any correct 3 readings 1A(4 Answer only) (4)
	4.1.2	Bus stopped ✓ 10✓	2RG (2)
	4.1.3	Yes✓ because 220 people visited her based on other visitors recommendations✓	2E (2)
	4.1.4	70 ÷ 380 ✓× 100✓ = 18,42✓%	2M (CA 4.1.1) 1A and rounding (3 Answer only) (3)
	4.1.5	Social networks✓ Fliers✓ (Accept any TWO appropriate methods)	2A (2)
4.2	4.2.1	75+460+185+90✓ = 810 ✓students	1M 1A(2 Answer only) (2)

4.2.2



1 mark – tittle (Any appropriate name)
1 mark – naming y-axis
1 mark – naming x-axis
4 marks 1 per each correct bar
1 mark – correct type of graph

(8)

4.3	4.3.1	Nokia✓	1RG (1)
	4.3.2	Samsung ✓and Huawei✓	2RG (2)
	4.3.3	LG% = 100% ✓- (20+9+20+14+27)✓ = 100 – 90✓ = 10✓%	2 RG 1M 1A(4 Answer only) (4)
	4.3.4	Nokia brand = 27% × 356✓ = 96,12✓ = 96 ✓Nokia cell phones	1M 1A 1R(3 Answer only) (3)
	4.3.5	The cellphones might have been: Expensive✓ Limited functionality✓ Not of good quality Old fashioned (Any TWO appropriate reasons)	 2J (2)
			[35]

TOTAL: 150

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